

The Division of Research

The Division of Research: Information Access and Independent-Book Capacity in LLM Investment Teams

Seven different LLMs reading the same 32 data fields produce about two independent portfolios. Giving each field to exactly one model nearly doubles independent breadth, with 68% fewer prompt tokens.

2.05 → 3.98

effective independent books, 7 LLM seats

shared access vs mixed split

+1.93

gain, 95% interval [+1.76, +2.09]

positive in all 34 month-ends

−68% / −52%

prompt tokens / model cost

mixed split vs shared access

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Seven AI analysts, two opinions

A one-minute read for someone new to the topic. Slides 03–09 give the evidence.

In plain terms

The problem Firms add several LLM "analysts" from different providers hoping for independent views, but feed all of them the same data. They end up proposing nearly the same trades and crowding into the same positions.

The idea Keep the models and the data fixed and change only who reads what: give each of 32 data fields to exactly one model ("mixed"), or one research domain per model ("coherent"), instead of everything to everyone ("shared").

The result Effective independent portfolios rose from 2.05 to 3.98 out of 7 seats, with 68% fewer prompt tokens, and the same ordering held in US, European and crypto markets.

MY ROLE · WHERE THIS COULD BE USED

My role Third author.

Use case Designing multi-agent LLM research desks; a cheap way to reduce crowded, correlated positions.

Terms used in this deck

SEAT

One LLM acting as an analyst; the paper uses seven, each from a different provider.

EFFECTIVE BOOKS (N_{eff})

How many truly independent portfolios a team really produces: 1 if every seat holds the same trade, 7 if all are independent.

ACCESS MATRIX

The table of which seat may read which data field; the only thing that changes between conditions.

DOLLAR-NEUTRAL BOOK

A long/short portfolio whose long and short sides are equal in dollar size.

SAME-SIDE OVERLAP

How often two seats hold the same stock in the same direction; a measure of crowding.

More models is not more independent views

Desks mix LLM providers but broadcast the same evidence to every seat.

Why it matters

- LLM analyst seats are cheap to add. Teams mix providers and roles while sending every seat the same feature store or memo.
- Outputs differ in wording but load the same trade: “adding models resembles adding assets without examining their covariance.”
- Prior work scores single-LLM quality or post-funding crowding, not how a fixed evidence set shapes the books a team proposes.
- Repeated same-side views share one liquidity channel, which means crowding and correlated-deleveraging risk.

WHAT THIS MEANS OUTSIDE THE PAPER

One variable changed same models, same data, different split of who reads what.

Measured on portfolios independent directions, not answer-level variety.

Generalizes to any multi-agent system that broadcasts one context to every agent.

Data & setup

MODELS

7 OpenRouter routes: OpenAI, Google, Meta, Alibaba, Mistral, DeepSeek, Anthropic

PERIOD

34 month-ends, Jul 2023–Apr 2026

UNIVERSE

14 U.S. large caps across 7 sectors

EVIDENCE

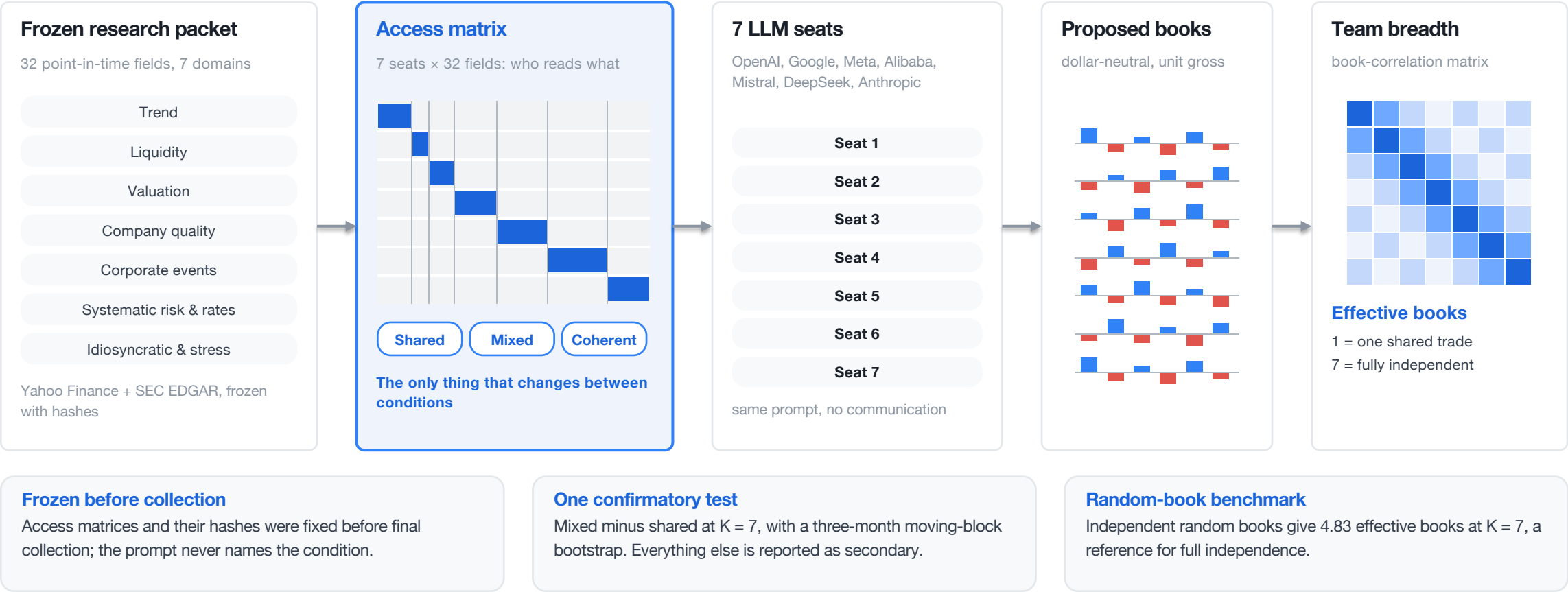
32 point-in-time fields from Yahoo Finance and SEC EDGAR, frozen with hashes

SCALE

13,090 archived books; 5,040 assignments averaged per date

Same models, same data; only access changes

A frozen 32-field packet reaches seven LLM seats through an access matrix, and the books they propose are compared.



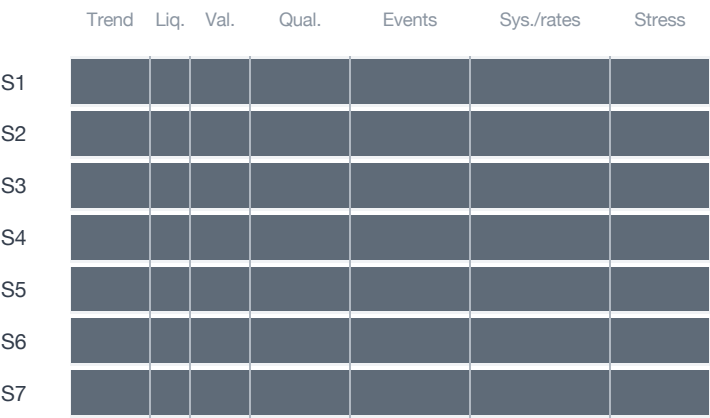
Redrawn and simplified from the paper's Figure 1.

Three ways to route the same 32 fields

Rows are seats and columns are data fields; a filled cell means that seat can read that field.

A. Shared

Every seat reads all 32 fields



N_eff 2.05 overlap 0.618 concentration 1.871

B. Mixed

Each field once; domains mixed



N_eff 3.98 overlap 0.381 concentration 0.818

C. Coherent

Each seat owns one research domain



N_eff 3.89 overlap 0.354 concentration 0.662

Effective books

$$N_{\text{eff}} = K / (1 + (K - 1) \cdot \rho^2)$$

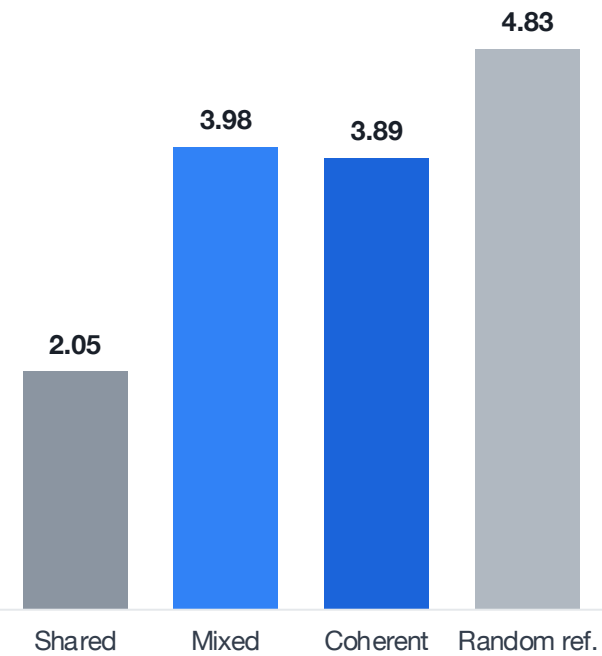
ρ^2 is the mean squared correlation between seat books. N_{eff} is 1 when every seat holds the same trade and K when every book is independent. Independent random books give 4.83 at $K = 7$.

Schematic redrawn from the paper’s Figure 1; exact cell assignments are in the paper. Values are 34-date means at $K = 7$.

Split access nearly doubles independent breadth

Seven seats, 34-month average. Split designs also cut duplicated demand in the proposed orders.

Effective books at K = 7



Proposed orders at K = 7

	Shared	Mixed	Coherent
Same-side overlap	0.618	0.381	0.354
External gross per \$100M	\$79.0M	\$45.7M	\$41.3M
Order concentration	1.871	0.818	0.662
Common-book correlation	0.725	0.215	0.147

What it shows

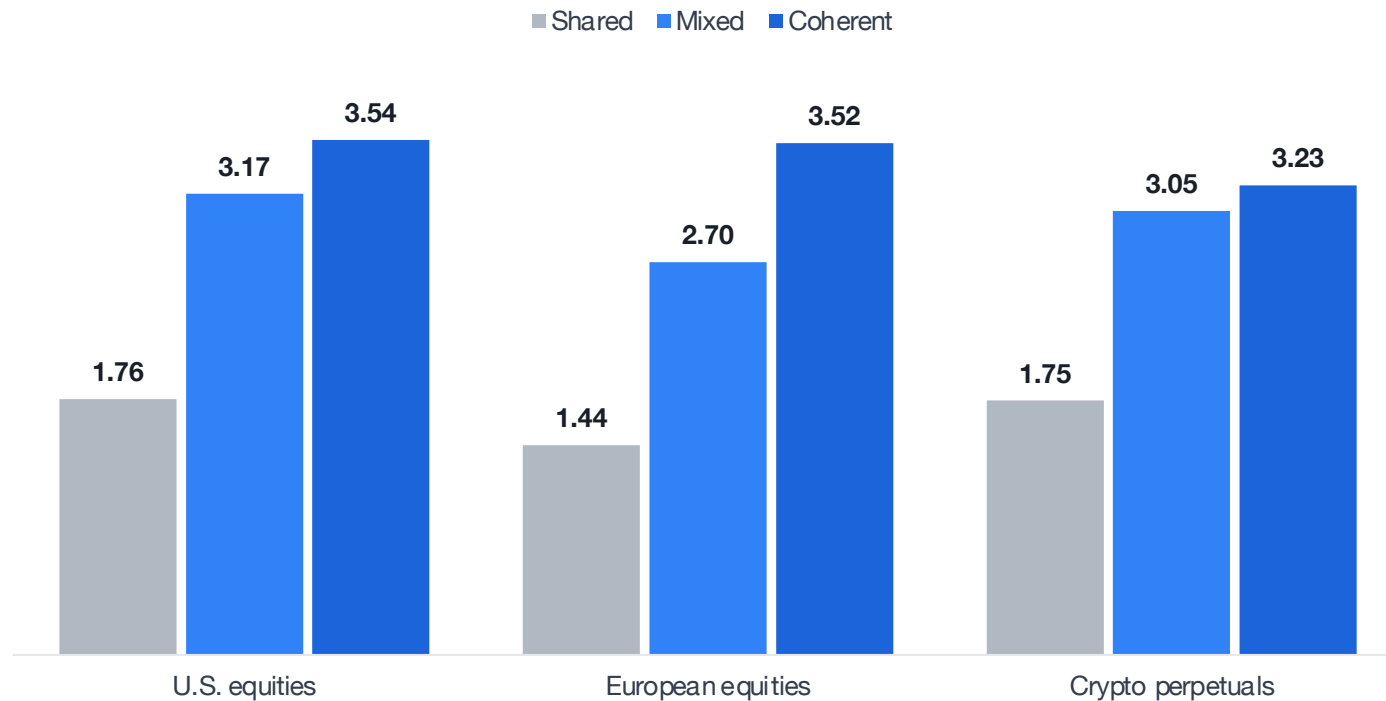
- The shared curve flattens fast: the 2nd seat adds 0.45 of an effective book, the 7th adds 0.05.
- Mixed reaches 82.4% of the random-book benchmark; shared reaches 42.5%.
- Coherent access keeps about the same breadth with the least duplicated demand.

Source: The Division of Research Table 2 and §5.1.

The ordering repeats across markets

Harmonized 12-date panel with the same 32 price-volume fields in each market.

Effective books at $K = 7$



The harmonized U.S. panel differs from the pooled U.S. panel on the previous slide.

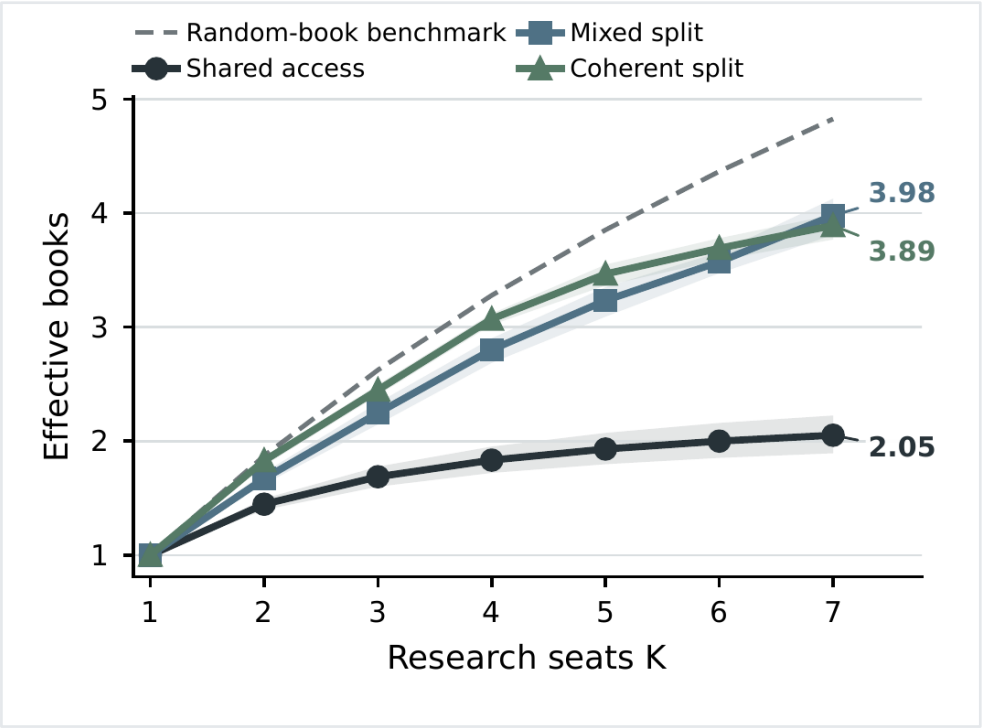
Source: The Division of Research Fig. 4, Table 4 and §5.4.

What it shows

- Split access beats shared access in every market. Here coherent tops mixed, the reverse of the pooled U.S. panel.
- Long-only transfer (gross): Sharpe 1.442 $\rightarrow$ 1.594 (mixed) / 1.566 (coherent); drawdown 16.57% $\rightarrow$ 10.18% / 9.46%; turnover 31.2% $\rightarrow$ 17.1% / 14.0%.
- 85.3–87.4% of the drop in team variance comes from lower cross-seat covariance.

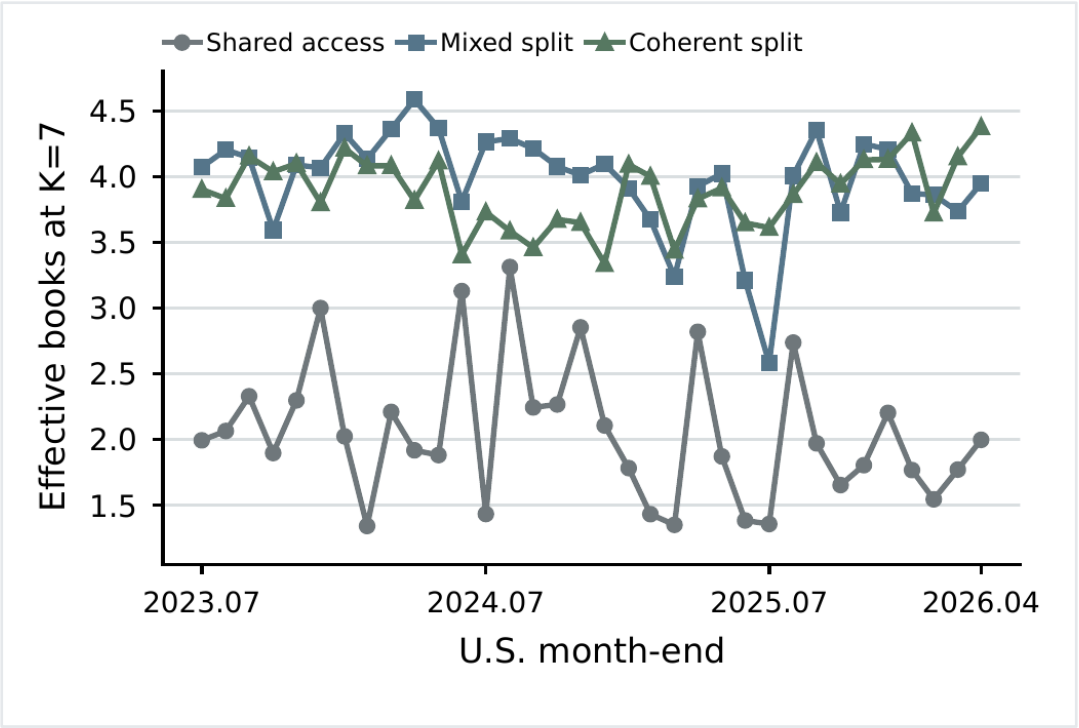
The effective-book curve, as published

Original figures from the submitted paper, shown as published. Numbers on earlier slides are read from its tables and text.



Paper Figure 2: Information access shifts the Effective-Book Curve

Shared access flattens near 2 as seats are added; split access keeps climbing toward the dashed random-book benchmark.



Paper Figure 3: The K = 7 access effect persists across all 34 U.S. month-ends

On every month-end, both split designs sit above shared access.

What held, and what the paper does not claim

Only mixed vs shared at $K = 7$ is confirmatory; the rest are stated as secondary checks.

Checks that held

- Held-out time: on 22 non-development dates the gains are +1.98 (mixed) and +1.87 (coherent), positive on all 22.
- After the latest model registration: +2.16 and +2.27 on 8 month-ends.
- Mixed beats shared at every team size from $K = 2$ to 7.
- Reruns at temperature 0.0 and 0.7, and a risk-space version: +0.94 [+0.71, +1.13].

What the paper does not claim

- It measures independent capacity, not alpha. In the long-only test split access had slightly lower gross returns (26.28% / 26.09% vs 27.49%), not significant.
- Packet scope and assignment change together, so no single mechanism is identified.
- One 14-stock universe and one model-vintage window.
- Orders are proposed demand only: no fills, implementation shortfall or realized impact.

Code (anonymous review repository): anonymous.4open.science/r/the-division-of-research-73C5